Symmetries in Physics — Exercise Sheet 8

Tensor products, selection rules and level splitting

Variant: Questions only

Exercise 1. The symmetry group D_4 of the square has the character table (Sheet 5, Exercise 2)

D_4	$1\,e$	$1\,r^2$	$2\,r$	$2\,s$	$2\,rs$
A_1	1	1	1	1	1
A_2	1	1	1	−1	−1
B_1	1	1	−1	1	−1
B_2	1	1	−1	−1	1
E	2	−2	0	0	0

The two-dimensional representation E of D_4 acts on vectors $\vec{a}$ in the plane of the square by the matrices

$$D(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the quarter turn and the reflection in the x-axis.

(i) Decompose $E \otimes E$ by characters.

(ii) Split it into S^2E and Λ^2E with the formula for the symmetric and antisymmetric squares (Lecture 8).

(iii) Let $\vec{a}$ and $\vec{b}$ transform by E. For each type found in (ii), give the combination of the products $a_\alpha b_\beta$ that carries it, as Lecture 8 does for C_{3v}. Why does the pair $(a_x b_x - a_y b_y,\ a_x b_y + a_y b_x)$, which carries E in C_{3v}, not stay together here?

Exercise 2. In a C_{3v}-symmetric system an operator (T_x, T_y) of type E acts between two doublet levels $\mathcal{L}_i$ and $\mathcal{L}_f$, both of type E. The bases $|i_\nu\rangle$, $|f_\mu\rangle$ of the levels and the pair (T_x, T_y) all transform by the matrices of Lecture 5, with hydrogen 1 on the y-axis:

$$D(C_3) = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \qquad D(\sigma_1) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

(i) Explain why the eight amplitudes $\langle f_\mu | T_\alpha | i_\nu \rangle$, $\mu, \alpha, \nu \in \{x, y\}$, are fixed by a single reduced matrix element.

(ii) Find the intertwiner $C\colon E \otimes E \to E$ by imposing $C\big(D(g) \otimes D(g)\big) = D(g)\,C$ for $g = \sigma_1$ and $g = C_3$.

(iii) Which amplitudes vanish, and how are the others related?

(iv) Show that the cubic $3x^2y - y^3$ is invariant under C_{3v}, and compare its coefficients with (iii).

Exercise 3. In a C_{3v}-symmetric system the electric dipole operator has the component z of type A_1 and the pair (x, y) of type E (Lecture 5).

(i) Use the selection rule of Lecture 8 to list the transitions between levels of types A_1, A_2 and E that light polarised along the axis (z) can cause, and those that light polarised in the plane (x, y) can cause.

(ii) Show that $A_1 \leftrightarrow A_2$ is forbidden for every polarisation.

(iii) What type must an operator have to connect an A_1 level with an A_2 level? Give a physical example.

Exercise 4 (formaldehyde in the infrared and in Raman scattering). Sheet 7, Exercise 2 placed formaldehyde H_2CO in the yz-plane, with the C–O bond along the z axis, and found $\Gamma_{\text{vib}} = 3A_1 \oplus 2B_1 \oplus B_2$ with the C_{2v} table

C_{2v}	E	C_2	$\sigma_v(yz)$	$\sigma'_v(xz)$	
A_1	1	1	1	1	z
A_2	1	1	−1	−1	
B_1	1	−1	1	−1	y
B_2	1	−1	−1	1	x

(i) Which fundamentals can be infrared active, and along which axis does the dipole of each oscillate? Use Lecture 8's argument: the matrix $L = \partial\vec{d}/\partial q$ at $q = 0$ satisfies $L\rho(g) = R(g)L$.

(ii) Compute the character of $S^2\Gamma_{\text{vec}}$ and decompose it. Which fundamentals can be Raman active?

(iii) Which type would be Raman active but silent in the infrared? Does formaldehyde have a mode of that type?

Exercise 5 (mutual exclusion rule).

(i) Let the point group of a molecule contain the inversion i. Show that $\rho(i) = \pm\mathbb{1}$ in every irreducible representation, so that each irreducible has a definite parity: g (*gerade*) for $\rho(i) = +\mathbb{1}$ and u (*ungerade*) for $\rho(i) = -\mathbb{1}$. Then show that a u mode can only be infrared active and a g mode only Raman active. Lecture 8 quotes this step for its mutual exclusion rule.

(ii) Boron trifluoride has $\Gamma_{\text{vib}} = A'_1 \oplus 2E' \oplus A''_2$ (Sheet 7, Exercise 5). Find Γ_{vec} and $S^2\Gamma_{\text{vec}}$. Decide which modes can be infrared active and which Raman active. Give the polarisation each infrared-active mode absorbs. Say why the mutual exclusion rule does not apply.

Data. The character table of D_{3h}, with classes of sizes 1, 2, 3, 1, 2, 3, is

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
A'_1	1	1	1	1	1	1
A'_2	1	1	−1	1	1	−1
E'	2	−1	0	2	−1	0
A''_1	1	1	1	−1	−1	−1
A''_2	1	1	−1	−1	−1	1
E''	2	−1	0	−2	1	0

The C_3 axis is z, perpendicular to the molecular plane σ_h.

Exercise 6. Stretch a square slightly along the x-axis into a rectangle. The symmetry drops from D_4 to its subgroup $D_2 = \{e, r^2, s, r^2s\}$, with s the reflection in the x-axis as in Exercise 1. Restrict each irreducible representation of D_4 (table in Exercise 1) to D_2. Show that a doublet E of the square generically splits into two nondegenerate levels of the rectangle.

Exercise 7 (a tetragonal crystal field). In Lecture 8's tutorial an octahedral site splits a d level into $E \oplus T_2$. Now stretch the octahedron slightly along the z axis.

(i) The rotations that survive form a group isomorphic to D_4, with r the quarter turn about z and s the half-turn about x. List its eight rotations, and find the class of O that each one belongs to. (In Lecture 8's table of O, $3C_2$ are the half-turns about the coordinate axes, and $6C_2'$ the half-turns about the axes through the midpoints of opposite edges.)

(ii) Restrict E and T_2 to D_4 and decompose (table in Exercise 1). How does the d level split, generically, under a small stretch? Which of the orbitals z^2, $x^2 - y^2$, xy, xz, yz goes with which type?

(iii) Back in the undistorted octahedron, use Lecture 8's rotation character formula to show that a p level does not split, and that an f level splits as $A_2 \oplus T_1 \oplus T_2$.